

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5108

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I A.Sai Joshitha(192424220)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Sai Joshitha

Question - 3

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-bystep.
- b. Explain how the repeating keyword influences each character of the ciphertext.

(a)Encryption Using Polyalphabetic Cipher

To encrypt the plaintext “HELLO” using the keyword “KEY” using a **Polyalphabetic Cipher**.

1.Given

Plaintext = HELLO

Key = KEY

The key is repeated to match the length of the plaintext.

Therefore:

Plaintext: H E L L O

Key: K E Y K E

4. Formula

The encryption formula is:

$$C=(P+K)\text{mod } 26$$

Where:

- **C** = Ciphertext value
- **P** = Plaintext value
- **K** = Key value
- **A = 0, B = 1, C = 2, ..., Z = 25**

5. Letter-to-Number Conversion

We use:

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Value	0	1	2	3	4	5	6	7	8	9	10	11	12
Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	13	14	15	16	17	18	19	20	21	22	23	24	25

Therefore:

H = 7, E = 4, L = 11, O = 14, K = 10, Y = 24

6. Step-by-Step Calculation

Character 1: H

Plaintext H = 7

Key K = 10

Formula:

$$C = (P + K) \bmod 26$$

Substitute:

$$C = (7 + 10) \bmod 26$$

$$C = 17 \bmod 26$$

$$C = 17$$

$$17 = \mathbf{R}$$

Therefore:

H → R

Character 2: E

Plaintext E = 4

Key E = 4

$$C=(4+4)\bmod 26$$

$$C=8$$

$$8 = \mathbf{I}$$

Therefore:

$$\mathbf{E} \rightarrow \mathbf{I}$$

Character 3: L

$$\text{Plaintext L} = 11$$

$$\text{Key Y} = 24$$

$$C=(11+24)\bmod 26$$

$$C=35\bmod 26$$

$$C=9$$

$$9 = \mathbf{J}$$

Therefore:

$$\mathbf{L} \rightarrow \mathbf{J}$$

Character 4: L

$$\text{Plaintext L} = 11$$

$$\text{Key K} = 10$$

$$C=(11+10)\bmod 26$$

$$C=21$$

$$21 = \mathbf{V}$$

Therefore:

$$\mathbf{L} \rightarrow \mathbf{V}$$

Character 5: O

$$\text{Plaintext O} = 14$$

Key E = 4

$C = (14 + 4) \bmod 26$

C=18

18 = S

Therefore:

O → S

7. Complete Calculation Table

Position	Plaintext	P	Key	K	Calculation	Ciphertext
1	H	7	K	10	$(7+10) \bmod 26 = 17$	R
2	E	4	E	4	$(4+4) \bmod 26 = 8$	I
3	L	11	Y	24	$(11+24) \bmod 26 = 9$	J
4	L	11	K	10	$(11+10) \bmod 26 = 21$	V
5	O	14	E	4	$(14+4) \bmod 26 = 18$	S

Final Answer: HELLO→RIJVS

Plaintext:HELLO

Keyword:KEY

Ciphertext: RIJVS

(b) How the Keyword Repeats

The keyword is: **KEY**

But the message has 5 letters.

Therefore:

KEY → KEYKE

So:

Plaintext : H E L L O

Key : K E Y K E

Ciphertext: R I J V S

The repeating keyword changes the shift applied to each character.

Result

The message **HELLO** is successfully encrypted as **RIJVS** using a **Polyalphabetic Cipher** with the keyword **KEY**.

Question – 5

5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as:

“WECRLTEERDSOEFEAOCAIVDEN”.

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Given

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

Number of levels = 3

(a) Determine the most probable original message by reversing the encryption process.

Step 1: Identify the method

transposition-based encryption method with three levels of arrangement.

For 3 levels, use the pattern:

1→2→3→2→1→2→3→2→...

This is the 3-rail zigzag arrangement.

Step 2: Count the ciphertext characters

WECRLTEERDSOEFEAOCAIVDEN

Number of characters:

25

Step 3: Find the number of characters in each level

Using the pattern:

1,2,3,2,1,2,3,2,...

For 25 characters:

Level 1 = 7 characters

Level 2 = 12 characters

Level 3 = 6 characters

Check:

$$7+12+6=25$$

Step 4: Divide the ciphertext

Ciphertext:WECRLTEERDSOEFEAOCAIVDEN

Level 1 — first 7 characters

WECRLTE

Level 2 — next 12 characters

ERDSOEFEAOC

Level 3 — remaining 6 characters

AIVDEN

So:

Level	Characters
Level 1	W E C R L T E
Level 2	E R D S O E E F E A O C
Level 3	A I V D E N

Step 5: Reconstruct using the pattern

Use:

1→2→3→2

Take letters in this order:

Position	Level	Letter
1	1	W
2	2	E
3	3	A
4	2	R
5	1	E
6	2	D
7	3	I
8	2	S

The complete decrypted message is:

WE ARE DISCOVERED FLEE AT ONCE

(b) Critically evaluate why such a method is vulnerable in modern communication systems.

A **transposition cipher** only changes the **position of the letters**. It does not strongly transform the actual characters.

Reasons for vulnerability:

1. Simple pattern

The three-level arrangement follows a predictable pattern:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 21$$

An attacker can identify or test this pattern.

2. Small number of levels

Only three levels are used, so there are limited arrangements to investigate.

3. Original letters remain unchanged

The letters in the plaintext are still present in the ciphertext; only their positions have been changed.

4. Can be attacked by trial

An attacker can try different transposition arrangements and look for meaningful words or sentences.

Therefore, the three-level transposition method is **weak for modern communication systems** because its arrangement is simple and predictable and it does not provide the strong confidentiality expected from modern cryptographic algorithms.

(c) Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Proposed enhancement:

Replace the three-level transposition cipher with a **modern authenticated encryption algorithm such as AES-GCM**.

Why AES-GCM?

It provides strong:

- **Confidentiality**
- **Data integrity**
- **Authentication**

Therefore, sensitive corporate messages can be protected much more effectively than with a simple transposition cipher.

The practical enhancement is to replace the legacy three-level transposition cipher with AES-GCM, a modern authenticated encryption technique, to provide stronger confidentiality and protection against unauthorized modification.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation